Kaustubh Dua

Software Developer · B.Tech. Electronics & Computer Engineering, JIIT Noida · Class of 2029

kaustubhdua1991@gmail.com · github.com/kaustubhdua · linkedin.com/in/kaustubh-dua-2b5a76416

PROFILE

Engineering student who ships full-stack products end to end: React or Next.js front ends, FastAPI and Supabase back ends, automated tests and CI. Careful about security, data boundaries and honest status labels. Open to software engineering internships in backend, full-stack and applied AI or data work.

EDUCATION

Jaypee Institute of Information Technology (JIIT), Noida
B.Tech., Electronics & Computer Engineering (ECM) · 2nd year of 4 · Expected graduation 2029

SKILLS

Languages Python · TypeScript · JavaScript · SQL · C++17 · Java
Build React · Next.js · FastAPI · Supabase · PostgreSQL / PostGIS · SQLAlchemy · Google OR-Tools · scikit-learn · PyTorch · OpenCV
Ship Git · GitHub Actions · pytest · Playwright · Appium · Docker · CodeQL

PROJECTS

NIRDHOOM Crop-residue coordination platform · prototype
React 19, Supabase, PostGIS, OR-Tools, PWA
- Built the full workflow from a farmer's booking to operator jobs, dispatch and a buyer-facing pool of **verified residue lots**.
- Kept service keys **server-side** behind Row Level Security; added SHA-256 evidence hashing and an offline upload queue.

github.com/kaustubhdua/nihdhoom · Live: nihdhoom-9qiu.vercel.app

Saathi Voice-first Hindi and Hinglish assistant · MVP
Next.js, FastAPI, speech APIs, GitHub Actions
- Made intent routing **deterministic** and tested it on a **120-case benchmark**, so the language model is never the source of truth.
- Attached a source and timestamp to every factual answer; added consent prompts and **human escalation**.

github.com/kaustubhdua/Saathi · Live: saathi-two-pi.vercel.app

GenomeGuard Governance layer for genomic data access · research prototype
Python, Starlette, NumPy, PostgreSQL, GitHub Actions
- Built consent-style policy checks, a disclosure-risk estimate and a **hash-chained audit log**; evaluated differencing and membership attacks on **synthetic data** and reported where the defences fail.

github.com/kaustubhdua/GenomeGuard

PatchPilot Pull-request risk triage · early stage
Python, FastAPI, pytest, Docker
- Wrote an analyser that turns a diff into **evidence-backed risk signals** and a 0-100 score, without ever running the code it reads.
- Hardened the API: input limits, strict schemas, rate limiting and a non-root Docker image.

github.com/kaustubhdua/PatchPilot

JIIT Youth Club website Public site for a campus club · live
React 19, Supabase, Playwright, CodeQL
- Built the site for events, clubs, gallery and recruitment, guarded by **browser tests and security scanning** on every change.

github.com/kaustubhdua/JYC-Website · Live: jyc-website-eight.vercel.app

CERTIFICATES AND HACKATHONS

- **JPMorgan Chase & Co. Software Engineering Job Simulation** (Forage, April to June 2026): Kafka and H2 integration, REST API integration and a REST controller on a Java and Spring Boot backend.
- **CS50 Introduction to Databases with SQL** (Harvard): schema design, indexing, joins and query optimisation.
- **OSDHACK'26**: Certificate of Merit for taking part; the team built Adapt AI.
- **ISRO Bharatiya Antariksh Hackathon 2026**: participated with AkashChalak, a satellite-assisted air-quality and hotspot prototype built on CPCB and NASA FIRMS data.

TEAMS, CLUBS AND VOLUNTEERING

- **JIIT Youth Club (JYC)**, Tech Vertical: member of the tech and management team.
- **AIML and AI-Tronics Club**, JIIT: member of the tech and management team.
- **National Service Scheme (NSS)**: volunteer on the cinematography and management teams.

TEAMWORK

Adapt AI (contributor) Local AI assistant that recommends an Ollama model for a machine's RAM and VRAM. Team project; my fork.